R/Pharma 2026 APAC

WinSual 溫究

Upstream Detection of CDISC Compliance Issues: Raw Data Quality Review with Local LLM and R

Winkle Lu

Clinical Statistical Programming · CDISC

R + Local LLM + Quarto

R/Pharma 2026 APAC

WinSual 溫究

Upstream Detection of CDISC Compliance Issues: Raw Data Quality Review with Local LLM and R

Winkle Lu

Clinical Statistical Programming · CDISC

R + Local LLM + Quarto

WinSual - Winkle blog


EDC
SDTM
ADaM
TLF
CSR



CDISC compliance checks run here

WinSual 溫究

The later an issue is found, the more it costs.

WinSual 溫究

We already work upstream.

So why do we still check downstream?

WinSual 溫究



This tool: Upstream Detection

Start from raw CRF data: map CRF field → SDTM variable → check collected values with CDISC / P21 rules

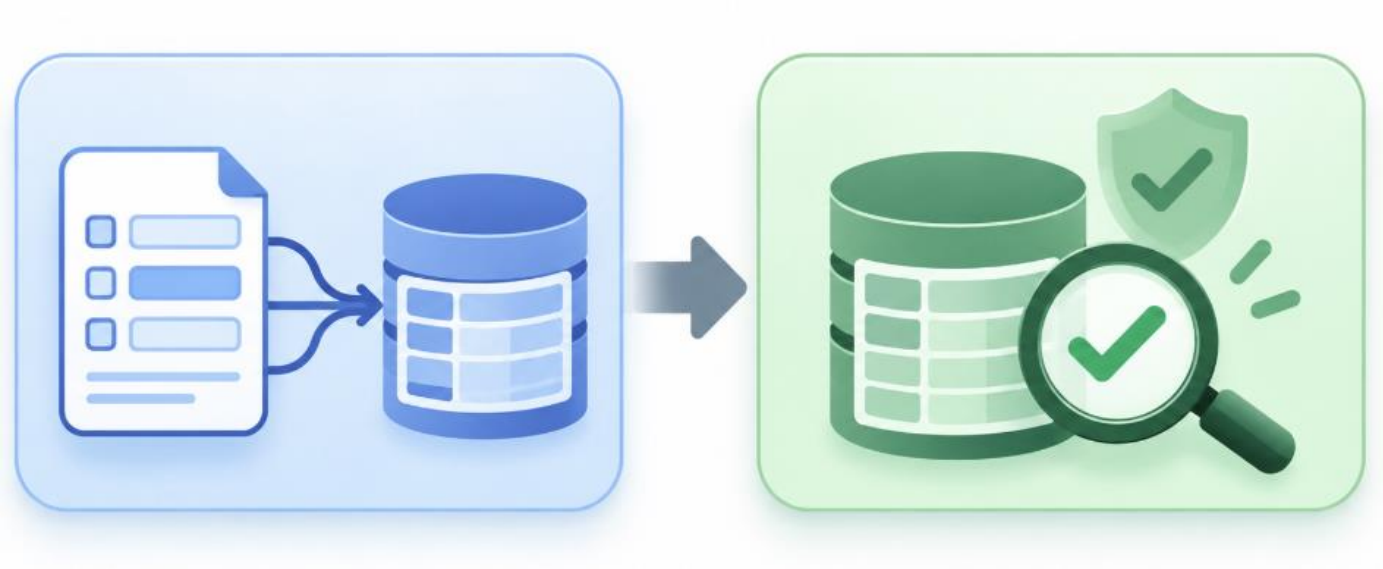
Programmers already write mapping and programs during data collection, so the checks can move earlier too.

EDC → SDTM → ADaM → TLF → CSR

This tool: Upstream Detection

Start from raw CRF data: map CRF field → SDTM variable → check collected values with CDISC / P21 rules

Programmers already write mapping and programs during data collection, so the checks can move earlier too.



Step 1: Mapping

Data-Fit Screening

rule out impossible variables

using collected values

Step 1: Mapping

Data-Fit Screening

rule out impossible variables using collected values

→

text2vec lexical similarity

Gemma 3 12B ranking

Gemma 3 4B answer + reason

Structured Chat reverse extraction

WinSual 溫究

Methods agree → take the answer.

Methods disagree → let a probability model decide.

WinSual 溫究

CRF label: "CTCAE Grade"

text2vec	**AETOXGR**
Gemma 3 12B	**AESEV**
Gemma 3 4B	**AESEV**
Structured Chat	**AELLT**

Jev picks

AETOXGR

0.92

Methods agree → take the answer. Methods disagree → let a probability model decide.

WinSual 溫究

Mock study · 45 subjects · ~6,000

DM, AE, VS, CM, EX

WinSual 溫究

48 of 57 scorable fields mapped correctly

84%

WinSual 溫究

Step 2: Generate the Report with Quarto



WinSual 溫究

- ✓ Death date before informed consent
- ✓ AE after death
- ✓ AE grade 6
- ✓ Height of 300 cm
- ✓ AE end date before start date

WinSual 溫究

5 / 5

300 cm passes every format check.
You need custom rules too.

Data issues and their locations across 5 SDTM domains (DM, AE, EX, CM, VS)

PUBLISHED
October 14, 2026

Contents

1. Raw data processed
2. Standards and rules referenced
3. Summary
4. Problem list
5. Problem details and locations
6. Search all issues

> **Note**
>
> **How to find where each problem is**
>
> 1. Section 4 "Problem list": every problem has an ID (e.g. P001) and shows the raw dataset, CRF field, number of records and subjects.
> 2. Click the ID to jump to Section 5: every record is listed with **subject, visit, row number in the raw data, record identifier** (e.g. `AESPID=3, AETERM=NAUSEA`) and the value.
> 3. Section 6 "Search all issues": type a subject ID, field name or value to filter.
>
> Raw data files: `Reference/CRF_data_injected/{raw dataset}.rds`. "Raw row" is the row number in that file (starting at 1). The raw DM dataset is processed in both the DM and VS rounds, so problems are grouped by "raw dataset (round)".

1. Raw data processed

The study's CRF exports were processed **before any SDTM dataset existed**: 10 raw datasets, 124 CRF fields across 5 SDTM domain rounds, 45 subjects.

SDTM domain (round)	Raw dataset	Records	Subjects	CRF fields mapped
DM	CO	45	45	4
DM	ICF	45	45	2
DM	DM	167	45	13
DM	TA	42	42	5
DM	DS	45	45	13
AE	AE	998	45	2
AE	AEAE	187	40	15
EX	DA	153	40	34
CM	CMCM	532	42	19
VS	DM	167	45	13
VS	VSVS	3,810	45	4

Raw data location: `Reference/CRF_data_injected/` — run IDs DM_20261013_234141, AE_20261013_234155, EX_20261013_234216, CM_20261013_234229, VS_20261013_234244.

2. Standards and rules referenced

Standard	Version
CDISC SDTMIG	v3.4
CDISC SDTM Controlled Terminology	2026-09-25
CDISC CDASHIG	v2.3
SDTM and SDTMIG Conformance Rules	v2.0
Pinnacle 21 validation rules	312 rules reviewed; 42 checkable at the CRF stage

By check

Check	Description	Problems	Error	Warning	Info	Total
CHK001	Date/time format (ISO 8601)	18	18	0	18	36
CHK002	Required variable missing or without source	7	2	6	0	8
CHK003	Date before informed consent	1	1	0	0	1
CHK004	Value not in CDISC Controlled Terminology	16	969	0	0	969
CHK005	Cross-form consistency	1	0	0	1	1
CHK006	Original value differs from its _FUL full value	8	0	227	0	227
CHK007	Start date after end date	1	1	0	0	1
CHK008	Event date after last participation date	7	89	0	0	89
CHK009	CTCAE grade outside 1-5	1	1	0	0	1
CHK010	Outside clinically plausible range	1	1	0	0	1
CHK011	Pinnacle 21 rule	12	424	328	0	752
MAP001	Mapping plausibility alert	5	0	0	5	5

Contents

1. Raw data processed
2. Standards and rules referenced
3. Summary
By check
4. Problem list
5. Problem details and locations
6. Search all issues

4. Problem list

Click an ID to see every record of that problem.

DM (DM round) — 4 problems

ID	CRF field	SDTM variable	Check	Rule	Severity	Records	Subjects	Values (examples)	Description
P001	SEX	SEX	CHK004 Value not in CDISC Controlled Terminology	CG0011	Error	45	45	FEMALE; MALE	Not in the SEX codelist; suggested standard term: F
P002	RACE	RACE	CHK004 Value not in CDISC Controlled Terminology	CG0020	Error	35	35	CAUCASIAN; INDIAN	Not in the RACE codelist (SDTMIG v3.4 standard values; codelist absent from the CT file); a recoding table is needed
P0	RACEAS	RACE	CHK004 Value not in CDISC	CG0020	Error	10	10	CHINA; OTHER,	Not in the RACE codelist (SDTMIG v3.4

WinSual 溫究

Some validation rules can be checked on raw CRF data

WinSual 溫究

Some validation rules can be checked on raw CRF data

Not a replacement for formal SDTM mapping or the validator

No change to the process — **just an early warning**

Serious issues go to **DM and PM early**

WinSual 溫究

LLM output is a suggestion, not an answer.

Data and standards constrain it · a probability model makes the call · humans review

WinSual 溫究

Full article by Winkle

Scan for the full write-up

WinSual 溫究

Backup slides

Q&A 1

Why local LLMs instead of GPT or Claude?

Mainly data privacy. The LLMs run locally, and the tie-breaker only sees metadata, so collected clinical values never leave the machine. And once data and standards constrain the problem, a local model is good enough — going from 37% to 70% came from the constraints, not a bigger model.

WinSual 溫究

Q&A 2

84% sounds good, but what about the other 16% — and the fields you didn't score?

The 84% is on the 57 fields where we had a reference answer from the annotated CRF. The nine still wrong are mostly conditional mappings, VS fields that need transposing, and a few whose reference answer deserves a second look. That's also why we keep human review in the loop.

WinSual 溫究

Q&A 3

About 2,000 issues — how many are false positives? Won't DM teams get alert fatigue?

In this mock study, only three were false alarms after we switched to Jev. And we group them by field: about 2,000 issues come from about 40 fields, so the team reviews 40 things, not 2,000.

WinSual 溫究

Q&A 4

Have you tried this on a real study, or other EDC systems?

Not yet — this mock study is adapted from a completed study. The next step is to run it on a live study, ideally with a different EDC system, and see how well the rules carry over.

Q&A 5

Why use Jev instead of a simple majority vote?

A majority vote tells you which answer won, but not how confident it is. Jev gives a probability, so we know which mappings a human should look at first. LLM self-reported confidence isn't reliable: the 12B model said 94% on average but was right 39% of the time on the disputed fields.

WinSual 溫究

Q&A 6

How is this different from just running P21 on a draft SDTM?

You need SDTM programs first to do that. This works on raw CRF data, before anyone has written a single SDTM program.

WinSual 溫究

Q&A 7

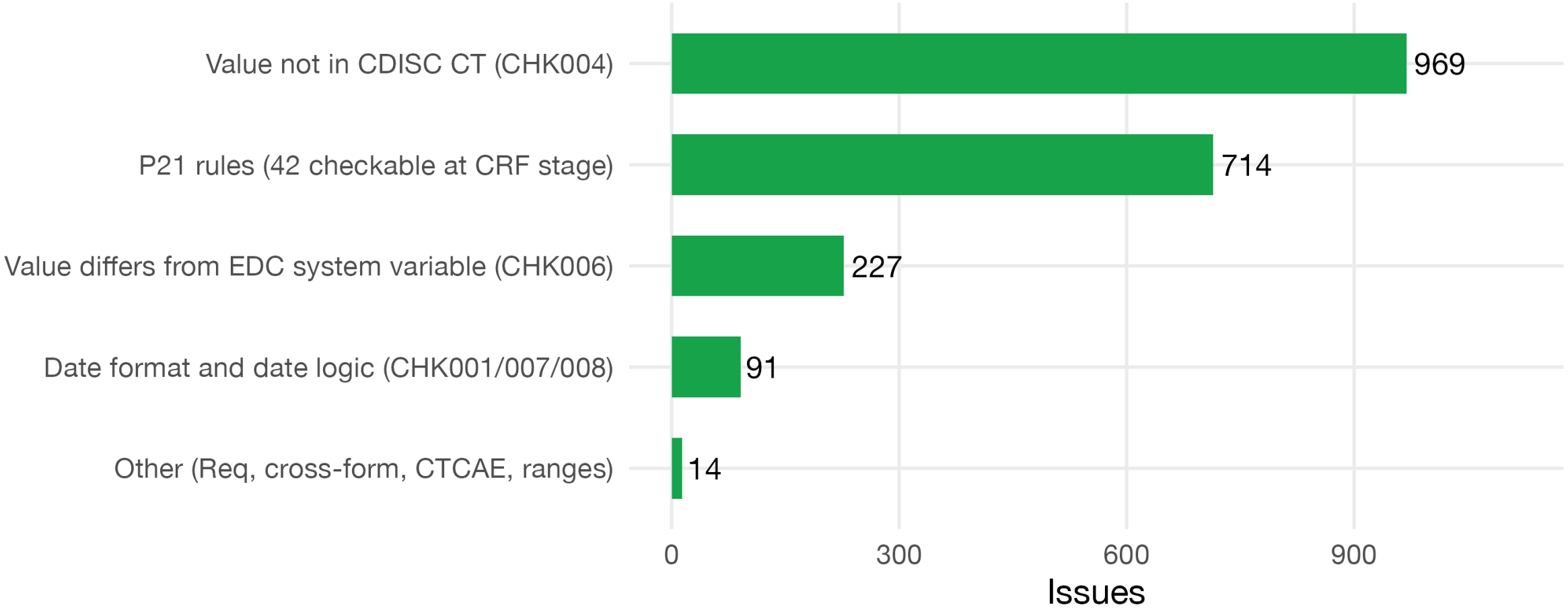
Is the code available?

Not at the moment. The QR code links to a full write-up on my blog that walks through the process, the methods and the results.

WinSual 溫究

2,015 data issues found before SDTM was built

From 40 CRF fields, covering all 45 subjects (official data, Oct 13 Jev run)



WinSual 溫究

Pipeline: from raw CRF data to a Quarto report

Input

Raw CRF data
10 raw datasets, 6,024 records

CRF spec + raw-to-domain map

CDISC documents
SDTMIG v3.4, SDTM CT, CDASHIG v2.3, Conformance Rules v2.0, P21 rules

→

Step one: Mapping

Step 00 | Prepare CRF fields, profile collected values

Step 01 | Load CDISC references

Step 02 | Data-Fit Screening + independent methods

Step 03 | Methods agree → accept; disagree → Jev arbitration

Step 04 | Update the custom dictionary

→

Step two: Validation

Step 05 | Check collected values via the mapping
ISO 8601, Req, CT, cross-form, date logic, CTCAE, plausible ranges, P21 rules

→

Output

Step 06 | Interactive Quarto HTML report

Step 07 | Run log

WinSual 溫究

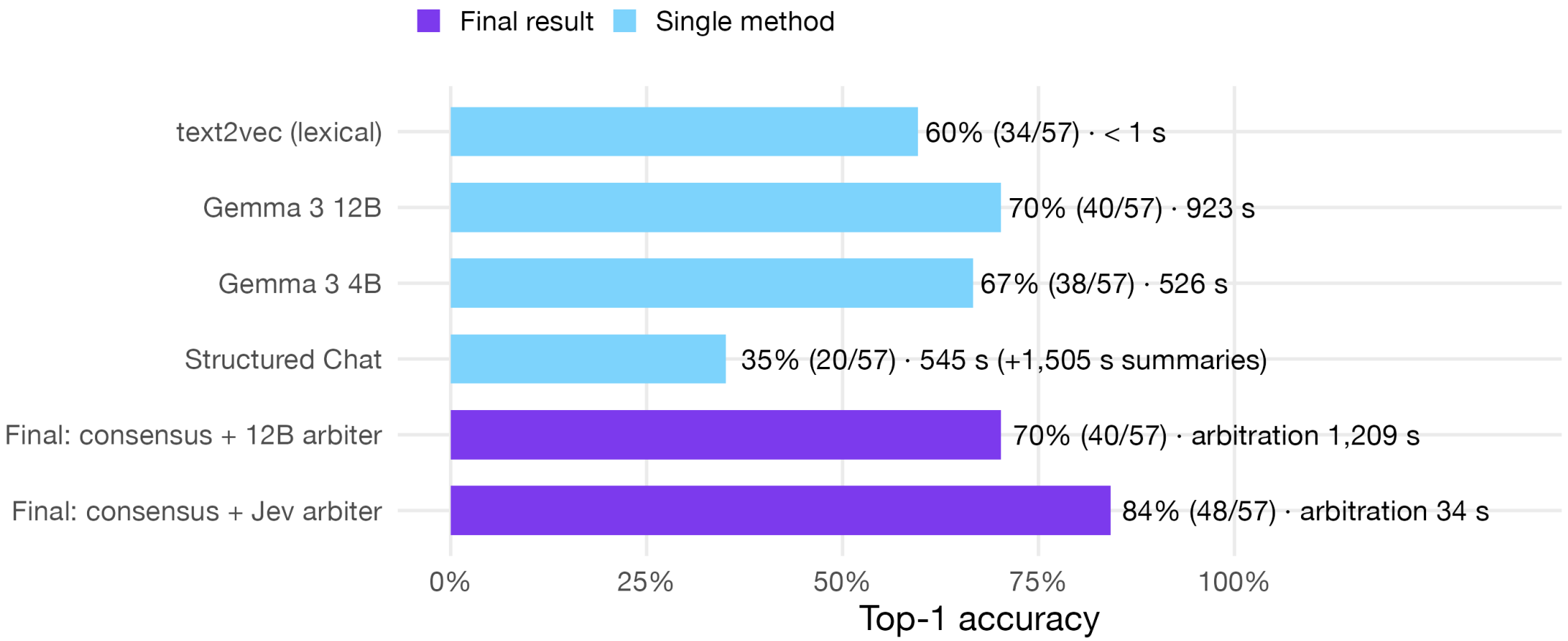
P026 | AESER | CHK011 Pinnacle 21 rule

Error | SDTM variable AESER | SD0009: No qualifiers set to 'Y', when AE is Serious | 15 records, 8 subjects

Subject	Visit	Raw row	Record identifier	Value	Message
SUBJ-002	AE	95	AESPID=1, AETERM=VOMITING, REPEATSN=1	YES	Serious AE, but the CRF has no SAE criteria fields (AESDTH, AESHOSP, etc.)
SUBJ-017	AE	15	AESPID=14, AETERM=CARDIAC TAMPONADE, REPEATSN=1	YES	Serious AE, but the CRF has no SAE criteria fields (AESDTH, AESHOSP, etc.)
SUBJ-017	AE	92	AESPID=13, AETERM=CHEST INFECTION, REPEATSN=1	YES	Serious AE, but the CRF has no SAE criteria fields (AESDTH, AESHOSP, etc.)

Mapping accuracy and run time by method

5 domains, 57 scorable fields. Data-Fit Screening recall: 96% (49/51), < 1 s

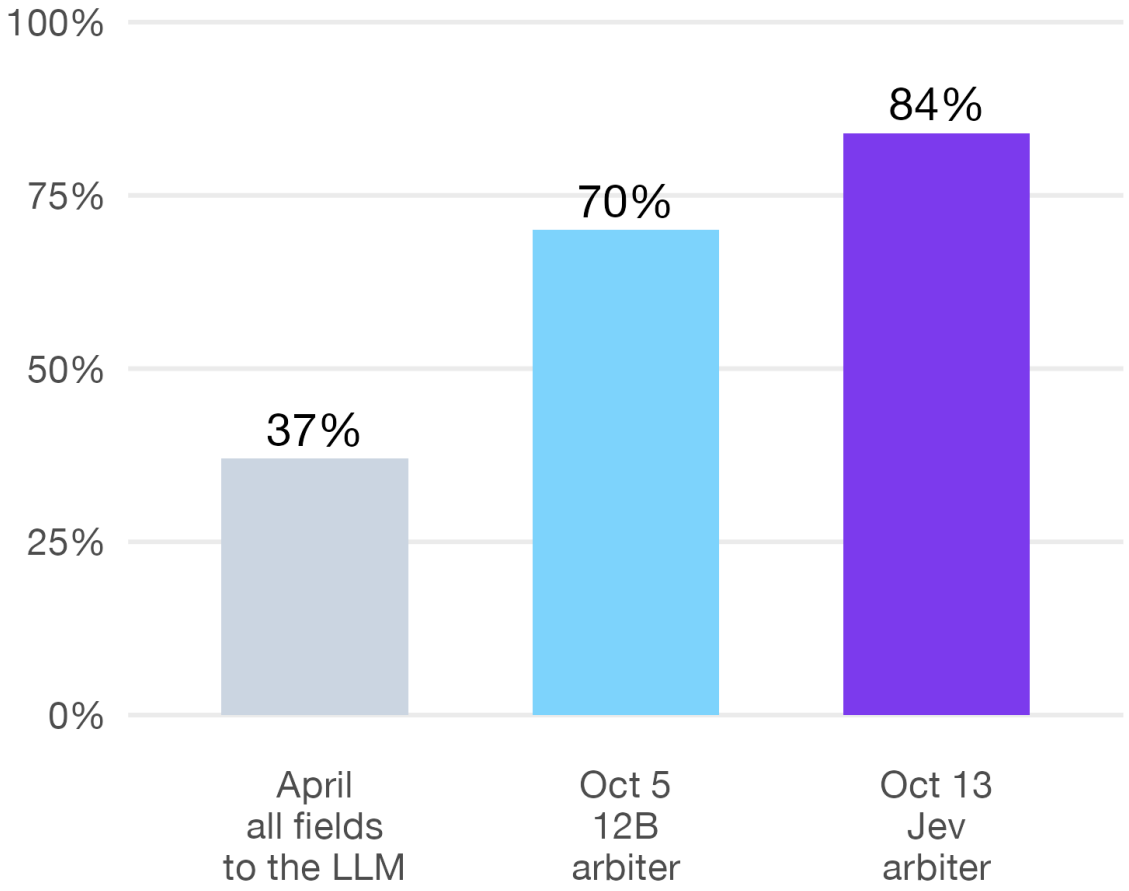


Times are machine time summed over 5 domains (local Mac, Ollama).
Jev takes about 0.5 s per call; the 12B arbiter about 19.8 s.

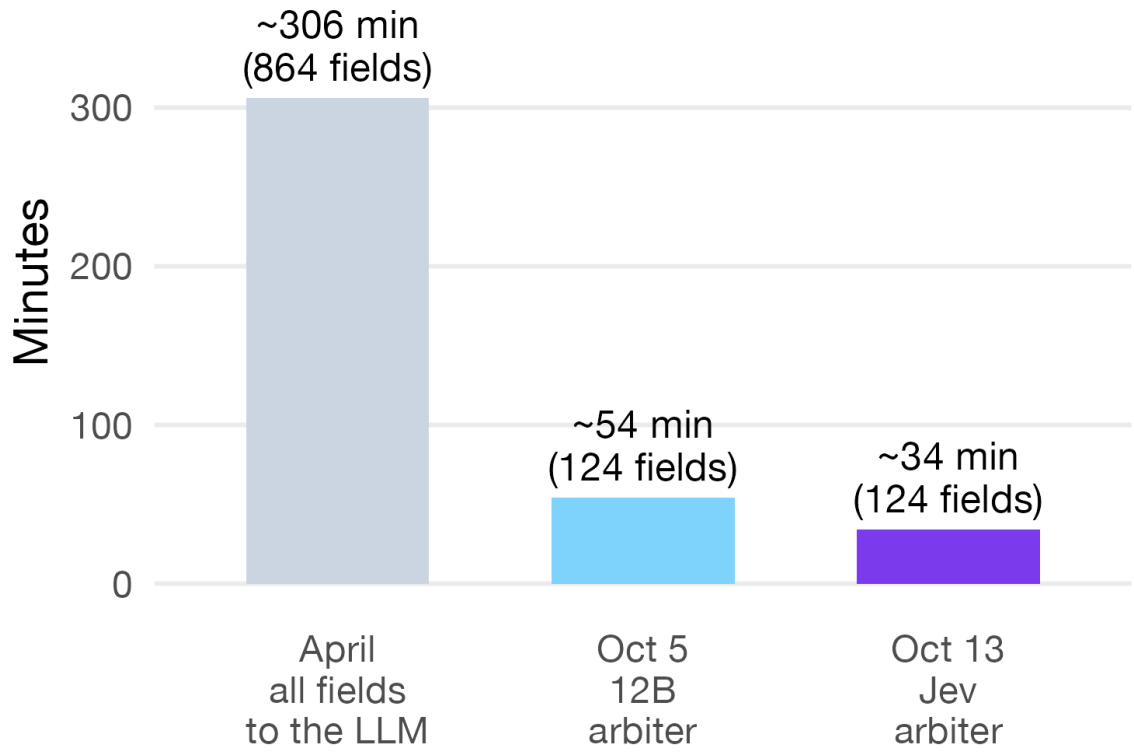
Everything to the LLM vs. constraining the LLM with data and standards first

(Oct 5 and Oct 13 both use screening + method consensus)

Mapping accuracy (same 57 fields)



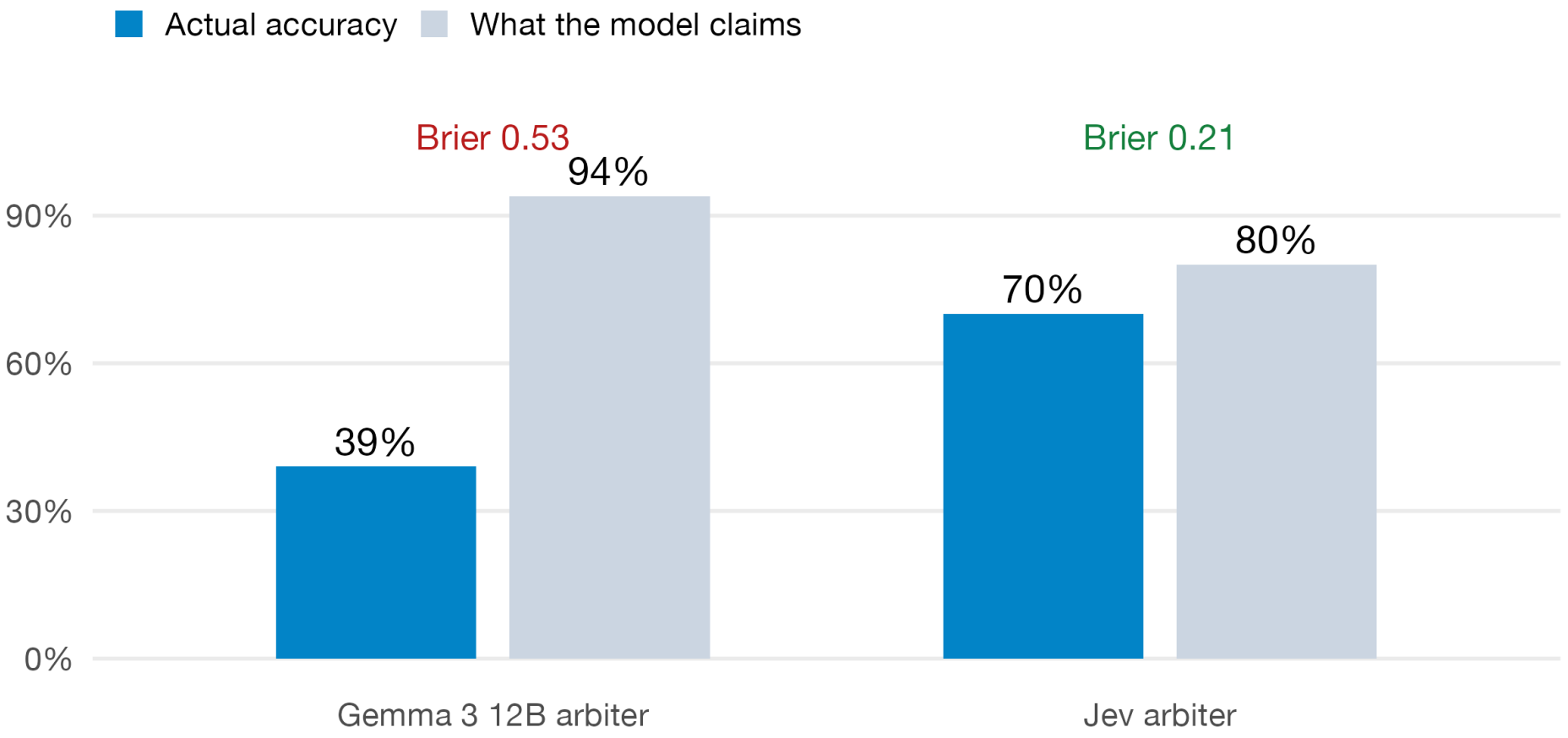
Machine time (same 10 raw datasets)



April time estimated from 13 hours / 2,200 fields; excludes one-time SDTM summaries (~25 min).

An LLM's stated confidence is not a probability

23 scorable fields where the methods disagreed



Jev figures are from the Oct 13 side experiment (74% in the official rerun).
Lower Brier score = more trustworthy probabilities.